MIL-R-10509/3H
22 May 1984
SUPERSEDING
MIL-R-10509/3G
9 May 1972

MILITARY SPECIFICATION

RESISTORS, FIXED, FILM (HIGH STABILITY),

STYLE RN70

* INACTIVE FOR DESIGN AFTER 9 MAY 1972 (SEE 6.2). USE MIL-R-55182/6. CHARACTERISTICS C, F, AND G, PREVIOUSLY INACTIVE FOR DESIGN AFTER 22 NOVEMBER 1968.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for style RN70, high-stability, film, fixed resistors. This style is available in all characteristics except B.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-10509 - Resistors, Fixed, Film (High Stability), General Specification for.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-10509, and as specified herein.

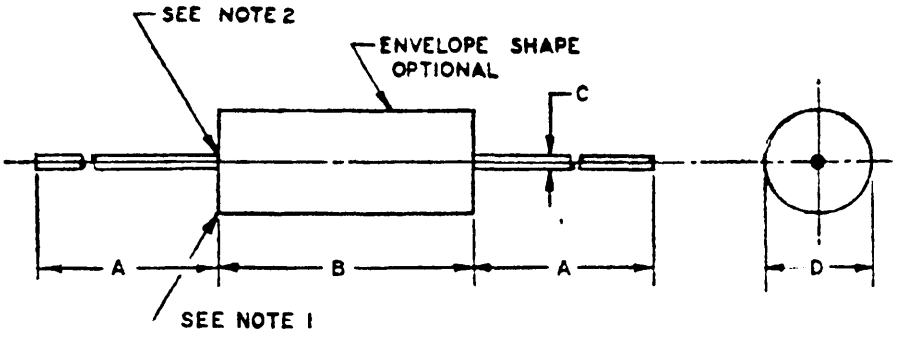
3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Power rating. The power rating shall be 1/2 watt. For characteristics D and F, the power rating shall be 3/4 watt. Characteristics D and F were formerly rated at 1 watt; however, the 3/4 watt is a direct replacement for the 1-watt unit.

3.4 Voltage rating. The maximum continuous working voltage shall not exceed 350 volts. For characteristics D and F, the maximum continuous working voltage shall not exceed 500 volts.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Electronics Research and Development Command, ATTN: DELET-R-S, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Symbol	Dimensions		
	Inches	Millimeters	
		Min	Max
A	1.500 ±.125	34.93	41.28
B	.750 (-.125, -.062)	17.48	22.23
C	.032 ±.002	0.76	0.86
D	.250 (-.078, -.031)	5.56	8.33

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The end of the body shall be that point at which the body diameter equals the nearest drill size larger than 150 percent of the nominal lead diameter, or 300 percent of the nominal lead diameter. The same drill size shall be used for each end of the body.
4. The leads shall be solderable to within .062 (1.58 mm) of the resistor body.

FIGURE 1. Style RN70 resistor.

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3.5 Resistance. Minimum and maximum resistance values shall be as follows:

Minimum resistance:		Maximum resistance:	
Characteristic:		Characteristic:	
D, G	-10.0 ohms.	D, G	2.49 megohms.
E, C, F	-24.9 ohms	E, C, F	-1.0 megohm.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-10509.

4.2 Terminal strength.

4.2.1 Direct load. Direct load shall be applied gradually until the load reaches 5 pounds.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements shall be in accordance with MIL-R-10509.

6. NOTES

6.1 The notes specified in MIL-R-10509 are applicable to this specification.

* 6.2 Substitution data. The resistors specified herein are not for use in design after 9 May 1972. They are authorized for use in design contracts effective prior to or on 9 May 1972. Existing equipment shall be supported by substitute resistors in MIL-R-55182/6 of the same resistance value, resistance tolerance, and characteristic performance of this specification regardless of failure rate designation. Resistors specified in MIL-R-55182/6 are preferred for design and regardless of failure rate designation can be used as substitutes for inactivated resistors of the same resistance value, tolerance, and characteristic performance of this specification. Characteristics C, F, and G were previously inactivated for design by revision F of this specification, dated 22 November 1968. Characteristic substitution shall be made as indicated in table I.

TABLE I. Characteristic substitution.

MIL-R-10509 Characteristic	Substitute MIL-R-55182 Characteristic
C	H
D	K
E	J
F	H
G	C

6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC, CG
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5905-1029-3)

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